

FUCHAI R180N DIESEL ENGINE

Test Report No	:	2013 - 112
Date of Test	:	September 12, 2013
Test Valid Until	:	September 12, 2018
Brand	:	FUCHAI
Model	:	R180N
Serial No.	:	194165
Type	:	Four stroke, water-cooled, single horizontal cylinder diesel engine
Manufacturer	:	Panyu Gaishan, Canshan District, Fuzhou, Fujian, China
Test Requested by	:	K-GEN INTERNATIONAL CORP. 93 Dapitan St., Sta Mesa Heights, Quezon City
Machine Classification	:	Commercial

SUMMARY

The FUCHAI R180N diesel engine was tested to establish its performance. It was tested at the full throttle speed setting pre-set by the manufacturer/ test applicant.

The Maximum Power developed by the engine was 5.27 kW at 2812 rpm engine Speed. At Maximum Power, the Torque on the output shaft of the engine was 1.83 kg-m with Fuel Consumption Rate of 5.03 L/h. The Noise Level measured at this condition was 88.6 db(A). The Maximum Torque developed was 2.04 kg-m occurred at 3.77 kW Output Power.

For the Continuous running test, the engine will be tested supposedly for five hours continuously. However, after half an hour of operation the engine stopped due to overheating. The test was stopped at this point.

SCOPE OF TEST

The engine was tested to establish its performance characteristics. The test was conducted on September 12, 2013 at the AMTEC Test Laboratory. It was evaluated on the following items:

- a. Suitability of operator's manual;
- b. Starting test;
- c. Varying load performance test; and
- d. Continuous running test

METHOD OF TEST

The engine was tested based on the Philippine Agricultural Engineering Standards, PAES 117:2000 Agricultural Machinery – Small Engines – Method of Test.

DESCRIPTION OF THE ENGINE

The FUCHAI Diesel Engine (**Figure 1**) had a Maximum Output Power of 5.67 kW (7.60 hp) at Rated Engine Shaft Speed of 2600 rpm and Continuous Output Power of 5.15 kW (6.90 hp) at Engine Shaft Speed of 2600 rpm. It was a four-stroke, water-cooled, single horizontal cylinder diesel engine. Fuel feed system was gravity type with the fuel tank located on top of the engine. It utilized an oil-bath type air cleaner. Lubrication system was force-feed lubricating type. Cooling was done by water circulating from cylinder jacket to the radiator using thermo-siphon principle. A fan driven by the pulley on the flywheel was used to cool the warm water in the radiator. It was started using manual cranking lever aided by a manually actuated decompression lever. A muffler was used in the exhaust system. **Table 1**, shows the detailed specifications of the engine.

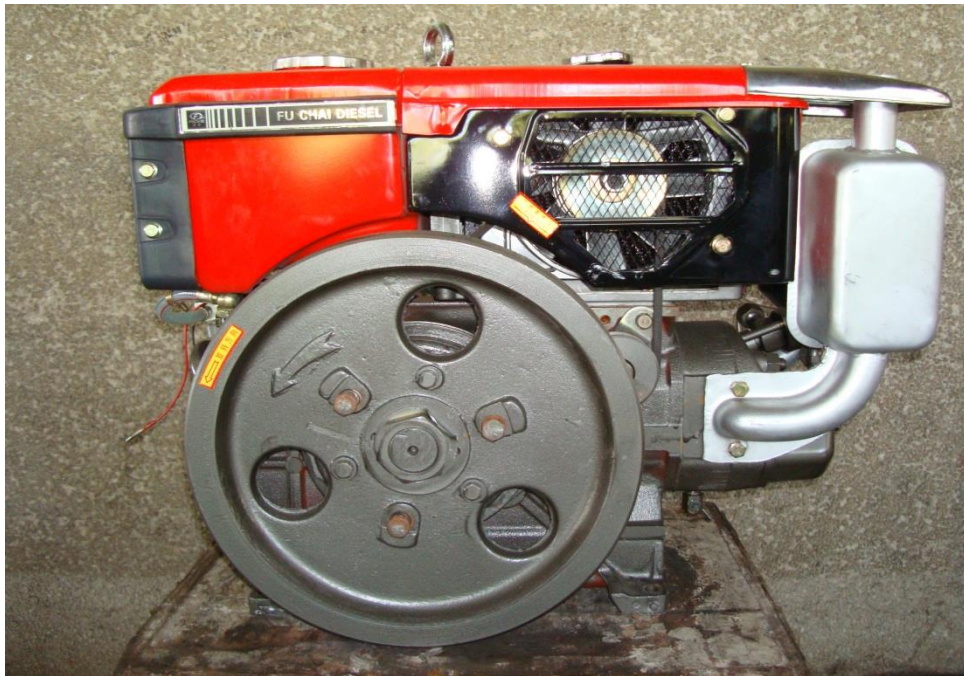


Figure 1. FUCHAI R180N Diesel Engine

ENGINE SPECIFICATIONS

Table 1. Detailed engine specifications of the FUCHAI R180N diesel engine

1	Dimensions and Weight		
1.1	Overall length (mm)	:	660
1.2	Overall width (mm)	:	340
1.3	Overall height (mm)	:	480
1.4	Weight (kg)	:	80.0
2	Engine specifications(Based on plate rating and operator's manual of the engine)		
2.1	Brand	:	FUCHAI
2.2	Model	:	R180N
2.3	Serial No.	:	194165
2.4	Make	:	Fuzhou, Fujian, China
2.5	Type	:	Four stroke, water-cooled single horizontal cylinder diesel engine
2.6	Rated Output, kW		
	Maximum Power	:	5.67 @ 2600 rpm
	Continuous Power	:	5.15 @ 2600 rpm
2.7	Bore x stroke (mm)	:	80 x 80
2.8	Total displacement volume (cc)	:	No data
2.9	Fuel system		
	2.9.1 Type of fuel feed	:	Gravity
	2.9.2 Fuel used	:	Diesel
2.10	Cooling system	:	Water-cooled
2.11	Lubricating system	:	Force-feed type
2.12	Starting system	:	Manual cranking lever
2.13	Air cleaner	:	Oil Bath type
2.14	Exhaust system	:	Muffler

OPERATOR'S MANUAL

No operator's manual was submitted with the engine.

RESULTS AND DISCUSSION

The FUCHAI R180N diesel engine was tested to establish its performance. The test was conducted by directly coupling the power output shaft of the engine to the electric dynamometer. The test set-up is shown in **Figure 2**.



Figure 2. Test set-up of the FUCHAI R180N Diesel Engine

A. STARTING TEST

Table 2 shows the result of the Starting test. The engine was started using the recommended starting procedure by the manufacturer. For cold starting, the engine was not operated for at least 12 hours and started, while for hot starting, the engine was operated for one (1) hour and then stopped and re-started. The number of attempts needed to start the engine was recorded in both tests.

During cold and hot starting, it only took a single attempt to start the engine. In general, the engine was easy to start.

STARTING TEST

Engine on Test : FUCHAI Diesel Engine
 Model : R180N
 Date of Test : September 12, 2013

Table 2. Results of starting test

ITEMS	COLD STARTING	HOT STARTING
1. Ambient temperature (°C)	28.3	
2. Relative Humidity (%)	91.4	
3. Engine oil temperature (°C)	28.6	58.0
4. No. of attempts to start the engine	1	1
5. Ease of starting	Easy	Easy

B. VARYING LOAD PERFORMANCE TEST

The engine mounted on the electric dynamometer was tested for varying load performance. Data were obtained at the recommended maximum no-load speed setting (Pre-set) of the engine and at prevailing ambient conditions during the time of test using locally available diesel fuel. **Table 3** and **Figure 3** show the results of the varying load performance test.

OUTPUT POWER. Based on the results of test, the Maximum Power developed by the engine, was 5.27 kW at 2812 rpm engine shaft speed. At Maximum Power, the Torque on the output shaft of the engine was 1.83 kg-m with Fuel Consumption Rate of 5.03 L/h. The Maximum Torque developed was 2.04 kg-m at an Engine Shaft Speed of 1803 rpm. The Actual Maximum Power was 93.0 % of the Rated Maximum Power (based on plate rating) of the engine.

FUEL CONSUMPTION. The Fuel Consumption of the engine was obtained using ONO SOKKI Digital Fuel Consumption Meter Model: FC-014/024 coupled to an ONO SOKKI Burette Stand Model: PP-523. The Maximum Fuel Consumption Rate of the engine was 5.03 L/h which occurred at the point of Maximum Output Power. The Specific Fuel Consumption at this point was 802.1 g/kW-h.

TEMPERATURES. Temperatures were measured using Type J thermocouples attached to a YEW Digital Thermometer type 2575. The Maximum Temperature of the engine oil was 90.0 °C which occurred at 1.92 kW Output Power of the engine and at 1093 rpm Engine Shaft Speed. For the exhaust air temperature at the outlet of the muffler, the Maximum Temperature was 581.5 °C which occurred at 4.96 kW Output Power and at 2598 rpm Engine Shaft Speed. For the cooling water at the radiator, the Maximum Temperature was 104.5 °C which occurred at 1.92 kW Output Power and at 1093 rpm Engine Shaft Speed.

NOISE LEVEL. Noise Level is the quantified degree of sound emitted by the engine during operation. It was measured using a SIMPSON Sound Level Meter Model: 884-2. The Noise Level of the engine was measured at 7.5 meters away from the muffler, 1.2 meters high from the ground and 45° from the engine shaft. The Maximum Noise Level measured was 88.9 db(A).

OBSERVATIONS. No breakdown and other malfunctioning occurred during the test.

VARYING LOAD PERFORMANCE TEST

Engine on Test : FUCHAI
 Model : R180N
 Date of Test : September 12, 2013

Test Conditions

Ambient Temperature (°C) : 28.3
 Relative humidity (%) : 91.4

Table 3. Results of varying load performance test¹

SHAFT TORQUE (kg-m)	ENGINE SHAFT SPEED (rpm)	OUTPUT POWER (kW)	FUEL CONSUMP- TION (L/h)	SPECIFIC FUEL CONSUMP- TION (g/kW-h)	TEMPERATURE (°C)			NOISE LEVEL [DB(A)]
					ENGINE OIL	COOLING WATER	EXHAUST AIR	
1.28	2977	3.91	1.47	315.1	58.0	62.0	294.5	87.3
1.83	2812	5.27	5.03	802.1	73.5	77.0	553.0	88.6
1.86	2598	4.96	4.77	807.2	78.5	85.5	581.5	88.9
1.87	2396	4.59	4.38	802.9	81.5	90.0	501.5	88.0
1.91	2206	4.32	4.04	785.2	88.0	96.5	493.5	86.0
1.92	2012	3.97	3.77	797.3	88.5	98.0	558.0	84.6
2.04	1803	3.77	3.35	746.2	89.0	102.0	566.5	83.0
1.82	1608	3.00	2.98	834.2	89.0	98.0	566.5	81.4
1.96	1376	2.77	2.43	737.0	89.5	103.5	522.0	77.4
1.72	1093	1.92	1.87	816.2	90.0	104.5	500.0	76.0
1.14	922	1.08	1.66	1292.8	89.5	102.5	433.5	72.4
0.71	793	0.58	1.48	2148.4	88.0	100.0	435.0	71.2

¹ Average of three test trials

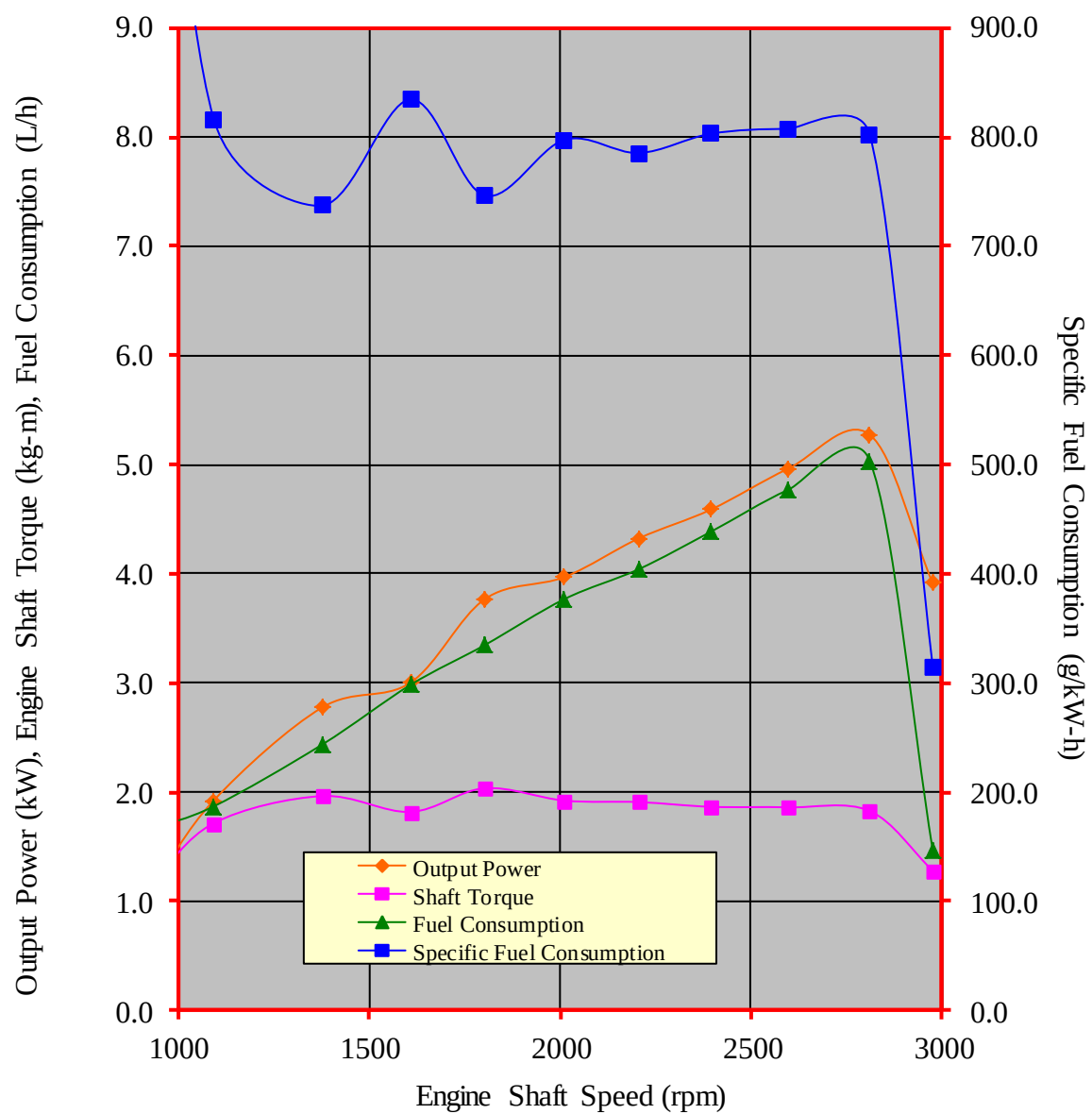


Figure 3. Performance characteristic curves of the FUCHAI R180N diesel engine

C. CONTINUOUS RUNNING TEST

For the Continuous running test, the engine will be tested supposedly for five hours continuously. However, after one hour of operation the engine stopped due to overheating. The test was stopped at this point.

OBSERVATIONS. The engine malfunctioned after one hour due to overheating during the test.

CONTINUOUS RUNNING TEST

Engine on Test : FUCHAI
 Model : R180N
 Date of Test : September 12, 2013
 Duration of Test (h) : 5

Test Conditions

Ambient Temperature (°C) : 30.3
 Relative humidity (%) : 84.0

Table 4. Results of continuous running test for one hour at 84.3 % of the Rated Maximum Power.

Items Measured	Average values
Engine Speed (rpm)	2615
Torque (kg-m)	1.78
Output Power (kW)	4.78
Fuel Consumption (L/h)	4.93
Specific Fuel Consumption (g/kW-h)	86.7
Temperature (°C) :	
a) Engine Oil	89.5
b) Exhaust Gas	541.5
c) Cooling Water	99.5
Noise Level [db(A)]	87.8

OBSERVATIONS/REMARKS: The engine stopped after half an hour of continuous running test.

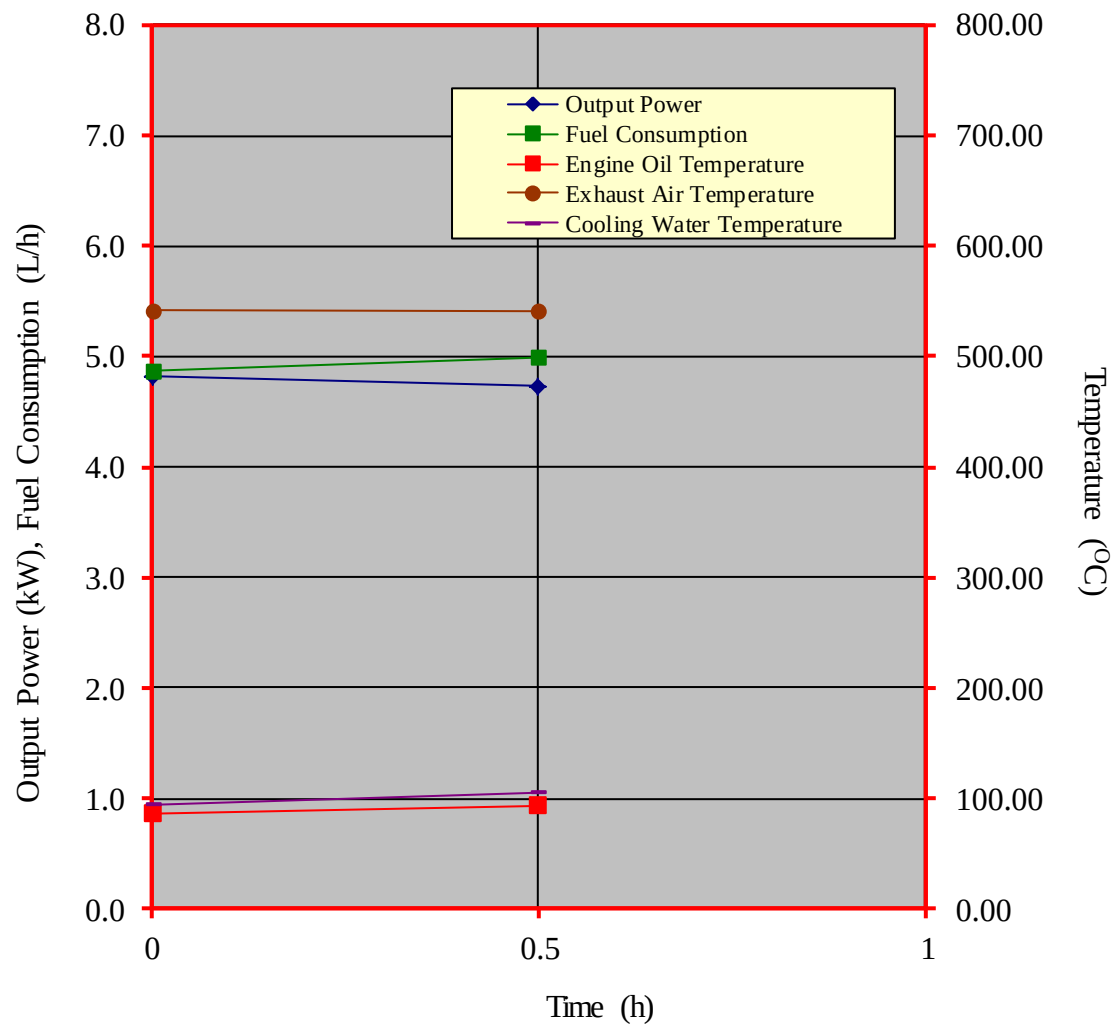


Figure 4. Performance characteristic curves during the continuous running test of the FUCHAI R180N diesel engine

DATA COMPUTATION

ANNEX. Formula used during calculation and testing

1. Varying Load Test

The engine was mounted on a dynamometer. The engine throttle was set at the maximum setting recommended by the manufacturer/test applicant. Data were gathered at different engine speeds starting from the maximum no-load speed down to the lowest speed. (The engine speed should not be lower than 600 rpm.) Data such as shaft torque; shaft speed; fuel consumption; temperatures of engine oil, exhaust air, and cooling water; vibration level at x, y, and z axis; and noise level were gathered at every speed setting. The following were computed:

- a. Output Power (kW), OP - The actual power at the output shaft of the engine is as given by equation 1.

$$OP = \frac{T \times N}{974} \quad , \text{ equation 1}$$

where:

OP = Output Power, kW

T = Torque, kg - m

N = Shaft Speed, rpm

$$\text{Percent power} = \frac{\text{Actual maximum power (kW) on test}}{\text{Rated maximum power (kW) found on the plate/broc hure}} \times 100$$

- b. Fuel Consumption (L/h) – It is the amount of fuel consumed by the engine per hour. This was obtained by volumetric method using ONO SOKKI Digital Fuel Consumption Meter (Model: FC-024) coupled to an ONO SOKKI Burette Stand (Model: PP-523).
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Annex, Con't.

- c. Specific Fuel Consumption (g/kW-h) - This is the amount of fuel consumed per unit of energy output of the engine. From test data, it was computed using equation 2.

$$SFC = \frac{FC \times D}{OP}, \quad \text{equation 2}$$

where:

SFC = Specific Fuel Consumption, g/kW - h

FC = Fuel Consumption, L/h

D = Density of Fuel used, g/L

- d. Temperature (°C) - These are the temperatures of the engine at different locations. They were measured using YOKOGAWA Digital Thermometer (Model: 2575) with Multi-Channel switch.

d.1 Engine Oil- the temperature of the engine oil at the crankcase during operation

d.2 Exhaust Air- the temperature of exhaust gas at the outlet of the muffler during operation

d.3 Cooling Water- the temperature of the water at the condenser during operation (if water cooled)

- e. Noise Level [db(A)]- The noise emitted by the engine 7.5 meters away from the muffler, 45° with respect to the axis of the engine shaft and the height of the microphone was 1.2 meters from the ground. **Table 5** shows the permissible exposure time for a certain level of sound emitted by the engine.

Table 5. Permissible exposure time for different sound levels

DURATION OF EXPOSURE PER DAY (h)	SOUND LEVEL [db(A)]
8	90
6	92
4	95
3	97
2	100
1.5	102
1	105
0.5	110
0.25 OR LESS	115

Based on "General Industry Guide for applying Safety and Health Standards",
29CFR 1910 OSHA, U.S. Dept. of Labor, 1973

Annex, Con't.

2. Continuous Running Test

This is the durability test for the engine. The engine was set to the recommended continuous power rating and at rated shaft speed found on the engine plate and/or brochure and was operated continuously for five hours. Data (same as in varying load test.) were gathered every 30 minutes. (If the engine has no recommended continuous power, it will be set at 80% of the rated maximum power and at rated shaft speed.) During the test, the shaft speed of the engine was not set lower than the rated. The average of the five-hour test data were reported as continuous test data of the engine. The average actual power of the engine divided by the rated maximum power found on the plate/brochure was reported as the percent continuous power attained by the engine.

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